

Dr. Frank Sposaro

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SUMMARY

Cloud Solution Architect and applied computer scientist with 20+ years of experience spanning Microsoft enterprise architecture, software engineering, AI-enabled systems, mobile computing, and applied research. My professional work focuses on helping organizations translate complex business goals into secure, scalable cloud and AI solutions using Microsoft Azure, Power Platform, Copilot Studio, and modern application architecture. I bring a rare combination of enterprise-scale architecture, hands-on engineering, teaching, research publication, and field curiosity, with a background that includes mobile medical monitoring, sensor-based systems, computer vision, low-code platforms, and AI-enabled application design.

I am now looking to apply those capabilities toward work with deeper personal meaning: using technology, AI, data, and field experience to support ecology, conservation, and environmental research. I am especially interested in opportunities involving scientific diving, marine field work, research expeditions, ecological monitoring, underwater observation, sensing, data collection, and exploration projects. My goal is to use my technical background for good by helping researchers, conservation teams, and mission-driven organizations build practical tools that improve how we understand, protect, and restore natural environments.

EXPERTISE

- Cloud Architecture, AI Systems, Applied Machine Learning, Low-Code Platforms, Enterprise ALM
- Microsoft Azure, Power Platform, Copilot Studio, Dataverse, Azure DevOps, Application Insights
- AI for Ecology, Conservation Technology, Marine Field Observation, Diving-Enabled Research Support
- Mobile Computing, Medical Monitoring Systems, Context-Aware Applications, Computer Vision, Sensor Systems
- Technical Strategy, Executive Workshops, Architecture Roadmaps, Teaching, Mentorship, Community Building

DIVING

- PADI Divemaster, 578633 2026
- PADI Open Water Scuba Instructor Candidate Est. August 2026
- GUE Technical Fundamentals Diver 2026
- GUE Cave and Technical Candidate Est. Fall 2026
- Field interests: marine ecology, underwater observation, conservation data collection, and diving-supported research

NONPROFIT

- Founder, The CESI Project* 2025–Present
Sposaro Conservation, Education, Sustainable Initiative www.cesi.earth
- Founded a mission-driven conservation and education initiative focused on connecting technology, environmental stewardship, and public engagement.
 - Developing the organization as a platform for conservation education, sustainability projects, ecological awareness, and applied technology for environmental good.
 - Exploring opportunities to combine AI, data collection, field research, diving, and community education in support of ecological and conservation outcomes.
 - Building CESI as a long-term personal fulfillment mission to apply technical, teaching, and field capabilities toward protecting and restoring natural environments.

EDUCATION

- Florida State University** 2004–2015
- Ph.D., Computer Science, 2015
 - M.S., Computer Science, 2012
 - B.S., Computer Science, 2008
 - Research areas: mobile medical monitoring, computer vision, sensory systems, and applied systems.

- PUBLICATIONS**
1. F. Sposaro, Z. Wang, and G. Tyson, "Programming Trends on Smartphones," *CSI Communications*, February 2013.
 2. M. Mitchell, F. Sposaro, A. Wang, and G. Tyson, "BEAT: Bio-Environmental Android Tracking," *IEEE Radio and Wireless Symposium*, pp. 402–405, January 2011.
 3. F. Sposaro, J. Danielson, and G. Tyson, "iWander: An Android Application for Dementia Patients," *Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, September 2010.
 4. F. Sposaro and G. Tyson, "iFall: A Cell-Phone Based Fall Detection and Warning System," April 2010.
 5. B. Fagin, F. Sposaro, and G. Tyson, "Indoor Tracking of Geriatric Patients Using Bluetooth," *First AMA IEEE Medical Technology Conference on Individualized Healthcare*, March 2010.
 6. F. Sposaro and G. Tyson, "Geriatric Medical Application Suite on a Sweet Phone," *First AMA IEEE Medical Technology Conference on Individualized Healthcare*, March 2010.
 7. F. Sposaro and G. Tyson, "iFall: An Android Application for Fall Monitoring and Response," *31st Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, September 2009.

- THESES**
- F. Sposaro, *Performance Gains Through Sensory Systems: A Dissertation*, Florida State University, 2015.
 - F. A. Sposaro, *A Geriatric Suite of Medical Applications for Android Powered Devices*, Florida State University, 2012.

PROFESSIONAL EXPERIENCE *Agentic Architect* 2025–2026
Professional Services, Microsoft

- Served as principal architect for AWS-to-Azure migration initiatives, defining multi-phase strategies for critical enterprise workloads.
- Authored architecture roadmaps aligning executive priorities with engineering execution and accelerating cloud-exit timelines.
- Designed secure landing zones and governance models ensuring compliance, scalability, and operational consistency.
- Led solution architecture for process digitization using low-code platforms and AI-enabled services.
- Established enterprise ALM strategy with CI/CD pipelines, environment isolation, and version control using Azure DevOps.
- Partnered directly with enterprise customers to influence architecture decisions and accelerate adoption of AI-enabled platforms.

Power Platform & Copilot Studio Specialist 2023–2025
Northeast Region, Microsoft

- Automated data ingestion, migrations, and CI/CD pipelines using Azure Functions, reducing manual effort and accelerating solution delivery.
- Led executive and technical workshops to align business objectives with solution architecture and delivery plans.
- Designed optimized data models and analytics solutions improving performance, reporting, and decision latency.
- Integrated third-party services and APIs into enterprise platforms to expand system capabilities and interoperability.
- Defined governance, security, and compliance policies enabling scalable low-code adoption across regulated environments.
- Diagnosed and resolved production issues through root-cause analysis, improving reliability and customer satisfaction.
- Designed and deployed AI-enhanced applications leveraging Python and Azure Cognitive Services.
- Implemented monitoring and observability for AI agents using Azure Monitor and Application Insights.

Adjunct Faculty 2023–2026
 Department of Computer Science, Florida State University

- Designed and delivered graduate-level coursework in Low-Code Application Development with emphasis on system design, automation, and AI integration.
- Taught Low-Code Application Development in Fall 2023, Spring 2025, and Spring 2026.

Power Platform Lead Architect 2020–2023
 Northeast Region, Microsoft

- Led enterprise digital transformation initiatives by orchestrating business stakeholders, engineering teams, and platform strategy.
- Facilitated executive workshops to define modernization roadmaps and translated requirements into scalable architectures.
- Delivered success metrics and adoption insights using Power BI and Azure analytics services.
- Migrated legacy systems to governed data platforms with RBAC, security controls, and compliance frameworks.
- Architected and delivered enterprise applications supporting mission-critical operations across multiple industries.
- Built conversational AI agents to support HR, training, and customer self-service use cases.
- Mentored architects and developers, fostering a community of practice around scalable low-code and AI solutions.
- Established ALM and governance standards adopted across multiple enterprise customers.

Cloud Solution Architect, Power Platform Developer 2018–2020
 Strategic Accounts, Financial Services, Microsoft

- Led cross-functional teams delivering AI-enabled and cloud-native solutions for strategic enterprise customers.
- Architected cloud solutions aligned to business objectives, regulatory requirements, and long-term scalability.
- Increased recurring revenue following deployment of audit and compliance platforms.
- Partnered with executive stakeholders to solidify business cases and prioritize transformation initiatives.
- Defined product roadmaps from prototype through production, aligning delivery with sales and customer success motions.
- Unified enterprise data using Dataverse, Azure Data Factory, and dataflows to support analytics and automation.
- Accelerated delivery of financial and tax solutions by optimizing agile processes and CI/CD pipelines.

Software Development Engineer 2013–2018
 Xbox Store, Microsoft, Redmond, WA

- Developed Python-based telemetry pipelines and executive reporting systems enabling data-driven roadmap decisions.
- Delivered high-impact e-commerce features used by millions, with emphasis on scalability, performance, and accessibility.
- Reduced cross-team dependencies by publishing reusable services and shared components.
- Mentored junior engineers and promoted agile best practices across teams.

CONSULTING *Power Platform Consultant, Owner* 2022–Present
 DrFrankDoes, LLC, Lakeland, FL

- Advises enterprise customers on Power Platform environment strategy, governance, application architecture, and lifecycle management.
- Developed a Power Pages property reporting application to support structured reporting and improved business visibility.
- Designed ALM processes for Power Platform solutions, including managed environments, deployment strategy, and release discipline.

- Created Power Automate calling flows for small-business operations and workflow automation.
- Released custom connectors to integrate existing APIs with Power Platform solutions.

EARLIER EXPERIENCE

Research Programmer and Technical Lead 2010–2013
Department of Computer Science, Florida State University, Tallahassee, FL

- Prototyped a medical monitoring framework using the Android platform, mobile sensors, and connected health devices.
- Developed automatic fall detection and alerting using triaxial accelerometers.
- Developed automatic wandering detection and alerting using GPS and contextual data for geriatric and dementia-related use cases.
- Built short- and long-term heart-rate monitoring capabilities using Bluetooth-connected sensors.
- Founded The Mobile Lab at FSU, a student and developer organization focused on mobile application development and applied mobile research.
- Branded and publicized the organization, recruited students, and built community participation around mobile development.
- Implemented and moderated a question-and-answer forum for mobile developers.
- Conducted interviews, performance reviews, and hiring activities for student talent.
- Managed a small group of student developers and coordinated applied research and software development activities.

Microsoft Student Partner 2012–2013
Florida State University, Tallahassee, FL

- Engaged with Microsoft developers, partners, MVPs, and campus communities to promote Microsoft technologies.
- Served as community outreach lead, brand ambassador, and event host.
- Drove more than 100 applications to publication on the Windows 8 Store through student engagement and technical support.

Android Developer 2012
iMobile3, Jacksonville, FL

- Automated Android build and installation processes, including lint and test reports.
- Designed a unit test framework for Android applications.
- Developed RoamPayX, a client-side payment application.

Software Engineer Intern 2011
Android Contacts Application, Google, Mountain View, CA

- Programmed user interface components for favorite contacts in the Android Phone and People applications.
- Designed code to support multiple screen sizes and device resolutions.
- Documented and fixed bugs for the Ice Cream Sandwich release.

Electronic Arts Campus Representative 2004–2011
Tri-Sqox Marketing, Redwood City, CA

- Managed public relations and promotional activities with clients and students at Florida State University.
- Planned, executed, and documented more than 150 promotional events.

Research Programmer 2008
Traffic and Criminal Software, FSU College of Engineering, Tallahassee, FL

- Prototyped law-statute matching software using C#.
- Created a GPS library with Visual Studio 2008 using C#.
- Automated build environment scripting using NAnt.

Software Engineer II 2007
ARINC, Oklahoma City, OK

- Simulated KC-135 MIL-STD-1553 aircraft data bus using Harris SecNet 11.
- Performed data analysis on RS-232 hardware GPS systems using XML and C++.

- Designed multiresolution, event-driven graphical user interfaces in VB.NET and C#.
- Automated user tasks for aircraft pilots using Windows and Unix scripts.

TEACHING

Adjunct Faculty, Course Designer and Remote Instructor 2023–2026
Department of Computer Science, Florida State University

- Low-Code Application Development
Semesters taught: Fall 2023, Spring 2025, Spring 2026
- Designed and remotely taught graduate-level coursework emphasizing system design, automation, AI integration, and applied low-code development.

Course Co-Founder, Primary Instructor 2010–2013
Department of Computer Science, Florida State University

- Mobile Programming: Windows 8 Application Development
Semesters taught: Spring 2013, Fall 2013
- Mobile Programming: Android Application Development
Semesters taught: Spring 2010, Fall 2011, Spring 2012

Teaching Assistant 2007–2010
Department of Computer Science, Florida State University

- C++, Computer Literacy, Computer Ethics, Data Structures, Web Programming

CERTS

